

APPENDIX A

THIS APPENDIX IS FOR THE PURPOSE OF PROVIDING GUIDANCE TO DESIGN ENGINEERS AND INCLUDES NO MANDATORY PROVISIONS.

10. DESIGN OF MECHANICAL SPRINGS

10.1 PURPOSE. The purpose of this appendix is to provide ready reference data to spring design information including data on materials, design, manufacturing processes and recommended tolerances. The information contained herein is intended to provide guidance to the designers of springs and not to impose restrictions in design.

10.2 SCOPE. The data contained in this appendix are sufficient for general spring design. This appendix is divided into three sections: Section I—MATERIALS Section II—DESIGN, and Section III—MANUFACTURE.

APPENDIX A, SECTION I

11. SPRING MATERIALS

11.1 GENERAL. More than forty different steel compositions, seven corrosion resisting steels and about twenty different non-ferrous compositions are used as spring materials. Each has some particular advantage over the others and the selection is based on seeking, in addition to other requirements, the best behavior in applications involving high stress, shock loads, elevated temperatures and corrosion resistance. All spring wire should be used in round cross-section whenever possible as it is more readily available, less expensive, usually has higher mechanical properties and is easier to fabricate. Steel wire has the longest fatigue life and can withstand the highest stresses. Copper-base alloys have the best electrical conductivity combined with good corrosion resistance. Corrosion resisting steels have good resistance to moderately elevated temperatures and to corrosion. Nickel-base alloys have the best resistance to elevated temperatures combined with excellent corrosion resistance. More specific data on these materials is covered in this section. The SAE and AISI references and the specifications of bar in the following paragraphs represent typical compositions of metal from which springs with the specified mechanical properties can be produced.

11.2 SIZES AVAILABLE. In the column headed Mechanical Properties in Paragraphs 12.2 through 17.3 are shown "Sizes Available", which lists the range of sizes usually available from warehouses or wire mills. For most materials both smaller and larger sizes can be obtained on special order. The sizes shown in Table I, Appendix A, Section I, should be used wherever possible.

12. HIGH CARBON STEELS

12.1 GENERAL. The most commonly used of all spring materials are the high carbon spring steels. These materials are used in Music-Wire, Hard-Drawn Wire, Oil Tempered Wire and Valve Spring Wire. All are readily available in a wide variety of sizes. These materials should be used in preference to other materials whenever the design requirements permit their use. They are not usually recommended for sub zero temperature applications.

MIL-STD-29A**12.2 MUSIC WIRE QQ-W-470, ASTM A228-51****Mechanical Properties:****Modulus;**

In tension

$E=30 \times 10^6$

In torsion

$G \text{ (Up to .100 in.)}$

$=12 \times 10^6$

$G \text{ (Over .100 in.)}$

$=11.75 \times 10^6$

Elastic Limit

Tension=65 to 75% of TS

Torsion=45 to 50% of TS

Density=.284 lb/in.³

Maximum elevated temperature

250° F

Sizes available: .004 to .180

Music wire is the highest quality cold drawn, high carbon wire in the ferrous group. It has exceptionally high tensile strength and uniform temper. It is recommended for small springs which are subjected to high stresses and suddenly applied loads. Only round sections have high tensile strengths. This material is not a true music wire if obtained in any sections other than round.

12.3 HARD DRAWN STEEL WIRE QQ-W-428, TYPE II; ASTM A227-47**Mechanical Properties:****Modulus;**

In tension

$E=30 \times 10^6$

In torsion

$G=11.5 \times 10^6$

Elastic Limit;

Tension=60 to 70% of TS

Torsion=45 to 55% of TS

Density=.284 lb/in.³

Maximum elevated temperature

250° F

Sizes Available: .028 to .625 in. dia

Hard drawn steel wire is a low cost spring steel which can be used for average stress applications where accuracy of wire diameter, spring diameter and loads are not required with great precision. Do not use in applications where long life is required. This material can be readily electroplated. Square sections are also obtainable but at reduced tensile strengths.

12.4 OIL TEMPERED STEEL WIRE QQ-W-428, TYPE I; ASTM A229-56**Mechanical Properties:****Modulus;**

In tension

$E=28.5 \times 10^6$

In torsion

$G=11.2 \times 10^6$

Oil-tempered steel wire is a general purpose spring steel of uniform quality and temper. Because of its surface condition this material is recommended for springs where: (1) the stress requirements are not too extreme, (2) the spring is not subjected to impact or shock loading, and (3) the spring index is 5 or greater. This material is widely used for me-

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Elastic Limit;
 Tension=80 to 90% of T S
 Torsion=40 to 50% of T S
 Rockwell Hardness=C45-50 less
 than .125 in dia
 C42-48 .126 in. to .250 in. dia
 C40-45 .251 in. to .500 in. dia

Density=.284 lb/in.³

Maximum elevated temperature
 350° F

Size Available: .082 to .500 in.
 dia and 1/32 to 1/2 in square sections
 in common fractions.

chanical products and machine tools in applications where the slight scale on the surface is not objectionable. The scale can be removed if necessary by shot blasting. Also obtainable in the annealed condition if desired.

12.5 CARBON-STEEL VALVE SPRING QUALITY WIRE ASTM A230-47**Mechanical Properties:**

Modulus;
 In tension
 $E=29.5 \times 10^6$
 In torsion
 $G=11.2 \times 10^6$

Elastic Limit;
 Tension=85 to 95% of T S
 Torsion=50 to 60% of T S

Rockwell Hardness=C45-50 for
 .125 in. diameter and under
 C42-48 above .125 in.

Density=.284 lb/in.³

Maximum elevated temperature
 350° F

Sizes Available:
 .093 to .250 in. dia

Because of the absence of scale on this material, it is suitable for use where flaking of scale would clog or otherwise injure moving parts. This material is recommended for springs of infinite life with an average stress range. This material is available in the tempered and annealed condition, but the tempered condition is most often used. The annealed condition should be used with subsequent hardening and tempering, for springs with a low spring index (less than 5) or where severe forming is required.

13. ALLOY STEEL WIRE AND BAR

13.1 GENERAL. Alloy spring steels are particularly useful in applications involving high stress and where shock or impact loadings occur. They can also withstand higher and lower temperature conditions than the carbon spring steels. All these alloys are generally used in the oil tempered or pre-hardened condition. They also are obtainable in the annealed condition for springs that have sharp bends or small spring indexes such as 5 or less.

MIL-STD-29A**13.2 CHROME-VANADIUM ALLOY STEEL WIRE & BAR ASTM A231-41 (Wire); QQ-S-624 FS 6150; A60-49 (Bars); ASTM A232-47 (Wire); QQ-W-412, Comp 1 (Wire)****Mechanical Properties:****Modulus;**

In tension

$E = 29.5 \times 10^6$

In torsion

$G = 11.2 \times 10^6$

Elastic Limit;

Tension = 88 to 93% of T S

Torsion = 65 to 75% of T S

Rockwell Hardness = C45-50

Density = .284 lb/in.³Maximum elevated temperature
425° F**Sizes Available:**

.032 to .375 in. dia in the tempered or annealed condition; 1/8 to 2 in. dia in bars for hot rolled springs. Square sections in common fractional sizes are also obtainable.

Chrome-Vanadium alloy steel wire will withstand higher stresses than Oil-tempered steel wire, but not as high as music wire. This material is recommended for springs which are subjected to shock by impact or suddenly applied loads, and for springs which are subjected to a large number of stress cycles. This material is frequently used for die springs. It is especially useful when annealed round wire is flattened into rectangular section with round edges. Die springs made from such sections are hardened and tempered after coiling and then shot-peened to increase their endurance limit. Hot rolled bars in heavy sections are also used for hot rolled springs and torsion bars.

13.3 SILICO-MANGANESE ALLOY STEEL BAR SAE 9260 & ASTM A59-49 (BAR)**Mechanical Properties:****Modulus;**

In tension

$E = 29.5 \times 10^6$

In torsion

$G = 11.2 \times 10^6$

Elastic Limit;

In tension = 78 to 86% of T S

In torsion = 55 to 65% of T S

Rockwell Hardness = C44 to 48

Density = .284 lb/in.³**Sizes Available:**

1/8 to 2 in. dia bars

This alloy steel has frequently been used as a less expensive substitute or alternate for chrome-vanadium. It does not have mechanical properties quite as high as other alloy steels. This alloy is obtainable in the oil-tempered, and in the annealed conditions in hot rolled bars. Heavy flat sections have been used for leaf springs and torsion bars.

MIL-STD-29A**13.4 CHROME-SILICON ALLOY STEEL WIRE QQ-W-412, COMP 2, TYPE II; ASTM A401-58****Mechanical Properties:**

Modulus;

In tension

 $E=29.5 \times 10^6$

In torsion

 $G=11.2 \times 10^6$

Elastic Limit;

In tension=88 to 93% of T S

In torsion=65 to 75% of T S

Rockwell Hardness=C50 to 53

Density=.284 lb/in.³

Maximum elevated temperature

475° F

Sizes Available:

.032 to .375 in. dia wire.

This alloy steel is especially suited for highly stressed springs subjected to shock or impact loading such as recoil springs in anti-aircraft guns and for moderately elevated temperatures. It may be obtained in the oil-tempered condition, but has most often been used in the annealed condition and then hardened and tempered to quite high hardnesses after coiling.

14. CORROSION RESISTING STEEL WIRE

14.1 GENERAL. Corrosion resisting steels are especially useful in applications involving corrosion and high temperatures. The 300 series "18-8" chromium-nickel austenitic types are usually used up to about $\frac{3}{16}$ inch diameter and the 400 series chromium martensitic types for larger sizes. The 300 series "18-8" types are hard-drawn to high tensile strengths and cannot be hardened by heat treatment. The 400 series martensitic types are usually used in the annealed condition and then hardened and tempered after coiling. The 300 series are non-magnetic in the fully annealed condition only—hard drawing to obtain spring qualities cause some magnetic ability which cannot be totally removed. The 400 series are magnetic and should not be used at sub zero temperatures. Passivating or immunizing by dipping in a 20 to 40% solution of nitric acid after fabrication is desirable for flat strip, but is frequently omitted on springs made of round wire.

14.2 CORROSION-RESISTING STEEL WIRE QQ-W-423, Composition FS 302, Condition B; AISI 302; SAE 30302; ASTM A313-55**Mechanical Properties:**

Modulus;

In tension

 $E=28 \times 10^6$

In torsion

 $G=10 \times 10^6$

Elastic Limit;

In tension=65 to 75% of T S

In torsion=45 to 55% of T S

This is a general purpose corrosion resisting material. Its spring properties are developed by cold working only. This material has higher tensile strengths than FS 304 and FS 316 but does not possess quite as good corrosion resisting properties. It is slightly magnetic in the spring temper, has low creep and resists relaxation at elevated temperatures. This material is also available in strip form for flat springs.

MIL-STD-29A**Rockwell Hardness=C42-47****Density=.288 lb/in.³****Maximum elevated temperature
550° F****Sizes Available:****.005 to .375 in. dia****Full Hard Temper up to .375 in. dia****Above .375 in. dia at lower hardness****14.3 CORROSION-RESISTING STEEL WIRE QQ-W-423, Composition FS 304, Condition B; AISI 304; SAE 30304****Mechanical Properties:****The same as FS 302, but the tensile strengths are about 5% lower.**

This material is quite similar to FS 302, has somewhat lower tensile strength, but has better bending properties and is an excellent alternate for most applications. The two types are often used interchangeably, however, FS 304 has somewhat better corrosion resistance than FS 302.

14.4 CORROSION-RESISTING STEEL WIRE QQ-W-423, Composition FS 316, Condition B; AISI 316; SAE 30316**Mechanical Properties:****The same as FS 302, but the tensile strengths are about 10 to 15% lower.**

This material has better corrosion resistance properties and is less magnetic than composition FS 302. Its mechanical properties at sub zero temperatures are better, but it cannot be used at stresses as high as FS 302. It has ability to take sharp bends due to the lower tensile strength.

14.5 CORROSION-RESISTING STEEL WIRE 17-7 PH**Mechanical Properties:****Modulus;****In tension** **$E = 29.5 \times 10^6$** **In torsion** **$G = 11 \times 10^6$** **Elastic Limit;****Tension = 75 to 80% of T S****Torsion = 55 to 60% of T S****Rockwell Hardness = C47-50
after hardening****Density = .277 lb/in.³****Sizes Available:****.030 to .375 in. dia**

This new type of corrosion resisting steel is of the precipitation hardening type. It is obtainable in the annealed condition, but is most often used in the cold worked condition and then precipitation hardened at 900° F for 1 hour. It then has tensile strengths higher than FS 302 and in some smaller sizes the tensile strength is equal to that of music wire.

MIL-STD-29A**14.6 CORROSION-RESISTING STEEL WIRE QQ-W-423, Composition FS 420; AISI 420; SAE 51420****Mechanical Properties:****Modulus;****In tension** **$E = 29 \times 10^6$** **In torsion** **$G = 11.20 \times 10^6$** **Elastic Limit;****Tension = 65 to 75% of T S****Torsion = 45 to 55% of T S****Rockwell Hardness = C46-51
after hardening****Density = .280 lb/in.³****Sizes Available:****.030 to .500 in. dia**

This corrosion resisting steel is usually used in diameters above $\frac{3}{16}$ inch, but occasionally it is used in smaller sizes such as .057 inch for recoil springs in rifles. It is always used in the annealed condition and then hardened and tempered after coiling. This material does not have corrosion resisting properties until after hardening. Clean bright surfaces provide best corrosion resistance and heat treating scale should be removed whenever possible.

14.7 CORROSION-RESISTING STEEL WIRE AISI 431; SAE 51431**Mechanical Properties:****Modulus;****In tension** **$E = 30 \times 10^6$** **$G = 11.5 \times 10^6$** **Elastic Limit;****Tension = 72 to 75% of T S****Torsion = 50 to 55% of T S****Rockwell Hardness = C47-51****Density = .280 lb/in.³****Sizes Available:****.050 to .312 in. dia**

This new type corrosion resisting steel is first hardened and tempered and then cold drawn. This combination produces bright clean wire with high tensile strengths nearly as high as music wire, but its corrosion resistance is not quite equal to FS 302. The hardened wire is magnetic and finds many uses for springs subjected to high stresses. Flat strip is also available.

15. COPPER-BASE ALLOYS

15.1 GENERAL. Copper-base alloys combine good electrical properties with excellent corrosion resistance. Although more expensive than steel they find many uses in electrical components and are excellent at sub zero temperature applications. All are non-magnetic.

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15.2 SPRING BRASS WIRE QQ-W-321, Composition B, Spring Temper; ASTM B134-52, Alloy 6; SAE 80A

Mechanical Properties:

Modulus;

In tension

$E = 15 \times 10^6$

In torsion

$G = 5.0 \times 10^6$

Elastic Limit;

In tension = 75 to 80% of T S

In torsion = 45 to 50% of T S

Rockwell Hardness = B89-95

Density = .308 lb/in.³

**Maximum elevated temperature
150° F**

Sizes Available:

.005 to .500 in. dia

Spring brass is recommended for use where high electrical conductivity, excellent corrosion resistance, ease of forming, low cost, and repeated flexure at low stress and low temperatures are required. Spring brass is non-magnetic. It is not recommended for applications where the stress is severe. It is also available in flat strip.

15.3 PHOSPHOR BRONZE WIRE QQ-W-401; ASTM B159-58, ALLOY A; SOE 81

Mechanical Properties:

Modulus;

In tension

$E = 15 \times 10^6$

In torsion

$G = 6.0 \times 10^6$

Elastic Limit;

Tension = 75 to 80% of T S

Torsion = 45 to 50% of T S

Rockwell Hardness = B90-97

Density = .320 lb/in.³

**Maximum elevated temperature
212° F**

Sizes Available:

.005 to .500 in. dia

Phosphor bronze is used for electrical conductivity, ability to resist corrosion, and non-magnetic properties. It has good bending and forming properties and the ability to withstand repeated flexures. This is the most popular of the copper-base non-ferrous alloys. It is also available in flat strip for electrical contact fingers. Superfine grain structure withstands quite high stresses and has long endurance to repeated flexure

15.4 BERYLLIUM COPPER WIRE QQ-C-530; ASTM B197-52

Mechanical Properties:

Modulus;

In tension

$E = 19 \times 10^6$

Beryllium copper is a precipitation hardening non-magnetic material with good electrical conductivity and corrosion resistance. It is non-magnetic and has high elastic and fatigue strength, and low drift and

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In tension

$$G = 7.3 \times 10^6$$

Elastic Limit;

Tension = 75% of TS

Torsion = 50 to 55% of TS

Rockwell Hardness = C37 to 41

Density = .298 lb/in.³

Maximum elevated temperature
300° F

Sizes Available:

.005 to .500 in. dia

hysteresis. This material can be severely formed in the annealed condition. Following forming, beryllium copper is precipitation hardened at 600° F for 2 hours. Pretempered wire is also useful for many applications where the stresses are not too high. Flat strip is also available.

16. NICKEL-BASE ALLOYS

16.1 GENERAL. Nickel-base alloys have excellent corrosion resistance combined with ability to withstand both elevated and below-zero temperature applications. Also, their non-magnetic characteristic is important for such devices as gyroscopes, chronoscopes and indicating instruments. These materials have high electrical resistance and should not be used for conductors of electrical current.

16.2 NICKEL-COPPER ALLOY, QQ-N-281, CLASS A, SPRING TEMPER

Mechanical Properties:

Modulus;

In tension

$$E = 26.0 \times 10^6$$

In torsion

$$G = 9.5 \times 10^6$$

Elastic Limit;

Tension = 65 to 70% of TS

Torsion = 38 to 42% of TS

TS = 165,000 up to .028 in. dia incl

160,000 over .028 in. to .057 in. dia incl

150,000 over .057 in. to .114 in. dia incl

140,000 over .114 in. to .312 in. dia incl

135,000 over .312 in. to .375 in. dia incl

130,000 over .375 in. to .500 in. dia incl

120,000 over .500 in. to .563 in. dia incl

Density = .319 lb/in.³

Maximum elevated temperature
400° F (450° F for short periods)

Sizes Available:

.005 to .250 in. dia full hard, over
.250 at lower hardness.

Monel is recommended for use at elevated temperatures where corrosive conditions exist. It retains its corrosive resistance properties at relatively high temperatures. This material has good resistance to sea water attack. It is a hard drawn wire and cannot be heat treated for hardening. Flat strip is also available. It is nearly non-magnetic.

MIL-STD-29A**16.3 NICKEL-COPPER-ALUMINUM ALLOY, WROUGHT, QQ-N-286, CLASS A****Mechanical Properties:****Modulus;**

In tension

$$E = 26.0 \times 10^6$$

In torsion

$$G = 9.5 \times 10^6$$

Elastic Limit;

Tension = 65 to 70% of T S

Torsion = 38 to 42% of T S

S = 180,000 up to .114 in dia incl

170,000 over .114 in. to .375 in. dia incl

160,000 over 375 in. to .563 in. dia incl

Density = .305 lb/in.³**Maximum elevated temperature**

450° F. (500° F. for short periods)

Sizes Available:

.005 to .563 in. dia for cold wound springs

3/8 in. dia and over for hot wound springs

K-Monel is a precipitation hardening alloy with excellent resistance to both corrosion and moderately elevated temperatures. It is usually used in larger sizes than Monel, as it can be formed in the cold drawn condition and then heat treated for hardening purposes. The mechanical properties are for the spring temper, age hardened condition. This alloy is non-magnetic.

16.4 WIRE: NICKEL-CHROMIUM-IRON ALLOY, QQ-W-390, CONDITION C**Mechanical Properties:****Modulus;**

In tension

$$E = 31 \times 10^6 \text{ at } 70^\circ \text{ F.}$$

$$29.5 \times 10^6 \text{ at } 400^\circ \text{ F.}$$

$$29.2 \times 10^6 \text{ at } 500^\circ \text{ F.}$$

$$28.7 \times 10^6 \text{ at } 600^\circ \text{ F.}$$

$$28.2 \times 10^6 \text{ at } 700^\circ \text{ F.}$$

$$28.0 \times 10^6 \text{ at } 750^\circ \text{ F.}$$

In torsion

$$G = 11.2 \times 10^6 \text{ at } 70^\circ \text{ F.}$$

$$10.8 \times 10^6 \text{ at } 400^\circ \text{ F.}$$

$$10.5 \times 10^6 \text{ at } 500^\circ \text{ F.}$$

$$10.2 \times 10^6 \text{ at } 600^\circ \text{ F.}$$

$$10.0 \times 10^6 \text{ at } 700^\circ \text{ F.}$$

$$9.8 \times 10^6 \text{ at } 750^\circ \text{ F.}$$

Elastic Limit;

In tension = 65 to 70% of T S

In torsion = 40 to 45% of T S

T S = 185,000 up to .057 in dia incl

175,000 over .057 in. to .114 in dia incl

170,000 over .114 in. to .229 in. dia incl

Inconel is recommended for springs requiring: (1) the retention of spring properties at relatively high temperatures, (2) corrosion resistance, (3) low magnetic permeability. This is one of the most popular of the nickel-base alloy group. It is cold drawn and cannot be hardened by heat treatment. Wire diameters up to 1/4 inch are most often used. This alloy is non-magnetic.

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165,000 over .229 in. to .329 in. dia incl
 160,000 over .329 in. to .375 in. dia incl
 155,000 over .375 in. to .500 in dia incl
 140,000 over .500 in. to .563 in. dia incl

Density = .304 lb/in.³

Maximum elevated temperature
 650° F. (750° F. for short periods)

Sizes Available:

.010 to .1875 in. dia full hard, over
 .1875 at lower hardness.

16.5 WIRE: NICKEL-ALLOY, SPRING, AMS-5698 for No. 1 temper; AMS-5699 for spring temper

Mechanical Properties:

Modulus;

In tension

$E = 31.00 \times 10^4$ at 70° F.
 28.1×10^4 at 700° F.
 27.7×10^4 at 800° F.
 27.2×10^4 at 900° F.
 26.7×10^4 at 1000° F.
 26.1×10^4 at 1100° F.
 25.5×10^4 at 1200° F.

In torsion

$G = 11.2 \times 10^4$ at 70° F.
 10.0×10^4 at 700° F.
 9.8×10^4 at 800° F.
 9.4×10^4 at 900° F.
 9.1×10^4 at 1000° F.
 8.8×10^4 at 1100° F.

Elastic Limit;

Tension = 65 to 70% of T S

Torsion = 40 to 45% of T S

T S = For No. 1 temper:

155,000 min up to .025 in. dia incl
 165,000 min over .025 to .468 in. dia
 For spring temper:
 220,000 min for .012 to .250 in. dia incl
 200,000 min over .250 to .418 in. dia incl

Density = .298 lb/in.³

Maximum elevated temperature
 1150 to 1200° F at low stresses

Inconel-X is a precipitation hardening material with high corrosion resistance and oxydization resistance at elevated temperatures. To make optimum use of this material, it is recommended that it be used in applications where the temperature is from 650 to 1150° F. For applications below 650° F the use of other materials is recommended. Neither cold setting nor shot peening should be specified on product drawing, when springs are intended to be operating within the recommended temperatures (650 to 1150° F) due to the adverse effect of pre-stressing on initial relaxation and the rate of relaxation under heat and stress. Large diameter bars can be hot coiled, and this is often done for $\frac{3}{8}$ in dia and larger sizes depending upon the spring index.

This non-magnetic alloy can also be used at sub-zero temperatures. The mechanical properties are for the tempers indicated after age hardening. For sub zero to 700° F applications, use spring temper stock and heat 1200° F for 4 hours, after coiling. For 700 to 1000° F applications, use No. 1 temper (or hot finished stock) and heat 1350° F for 16 hours, after coiling. For 1000 to 1200° F applications, use spring temper stock and heat 2100° F for 2 hours, air cool, then 1550° F for 24 hours, air cool and 1300° F for 20 hours and air cool, after coiling.

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.012 to .468 in. dia incl No. 1 temper

.012 to .418 in. dia incl spring temper

.375 in. dia and over, hot finished (for hot wound springs)

17. STEEL STRIP, HIGH CARBON

17.1 GENERAL. Several types of flat cold rolled steel strip with different ranges of carbon content, tempers, finishes and edges are obtainable, but only two types are readily available. These two types are used for over 95 per cent of all applications requiring steel strip. Thin sections of high hardness (over Rockwell "C" 43) are highly susceptible to hydrogen embrittlement resulting from electroplating or pickling operations. These materials are used for mainsprings in clocks and timing devices; for constant force springs, clips and stampings. For general information see Spec. MIL-S-17919.

17.2 STEEL STRIP, HIGH CARBON SAE 1074; MIL-S-17919 MATL. No. 1 & 4**Mechanical Properties:****Modulus;**

In tension

 $E=30 \times 10^6$

In torsion

 $G=11.5 \times 10^6$

T S=244,000 to 305,000 for sizes under .032 in.

214,000 to 240,000 for sizes over .032 in.

Elastic Limit;

Tension=85 to 90% of T S

Torsion=65 to 75% of T S

Rockwell Hardness=C46 to 49

Density=.284 lb/in.³

Maximum Elevated Temperature 350° F

Sizes Available:

.005 to .062 in. thickness,

others on special order.

This type of material is usually called "cold rolled blue tempered spring steel". It is slit to desired widths and the edges can be squared or rounded as desired. The material obtained in the hardened condition is widely used for flat springs, spiral, clock and motor springs. It is also available in the annealed condition for use in automatic equipment and where severe forming operations are required, after which it is hardened and tempered.

17.3 STEEL STRIP, HIGH CARBON SAE 1095; MIL-S-17919 MATL. No. 1 & 2**Mechanical Properties:**

The same as SAE 1074 but the tensile strengths are about 5 per cent higher.

Rockwell Hardness=C48 to 51

This type of material, usually called "cold rolled blue tempered and polished clock spring steel", is widely used in clocks and motor springs because it can withstand higher stresses than SAE 1074. It is slit to widths desired and the edges are usually made square or rounded. It is generally obtained in the hardened condition, but annealed material can be obtained.

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TABLE I. Preferred sizes for spring materials wire diameters, strip thickness and bars

Spring steels			Corrosion resisting		Copper and nickel alloys	
Music wire	High carbon & alloy steels	Valve spring quality steels	"18-18" chrome nickel austenitic 300 series	Straight chrome martensitic 400 series	Spring quality brass phosphor bronze beryllium cop. monel & Inconel	K monel & Inconel X
.004	.032	.092	.004	Same as high carbon & alloy steels, Col. 2	.010	.125
.006	.035	.105	.006		.012	.156
.008	.041	.125	.008		.014	.162
.010	.047	.135	.010		.016	.188
.012	.064	.148	.012		.018	.250
.014	.063	.156	.014	Same as high carbon & alloy steels, Col. 2	.020	.313
.016	.072	.162	.020		.025	.375
.018	.080	.177	.026		.032	.475
.020	.092	.188	.032		.036	.500
.022	.106	.192	.042		.040	.563
.024	.125	.207	.048		.045	.688
.026	.135	.218	.054		.051	.750
.028	.148	.225	.063		.057	.875
.032	.156	.244	.072		.064	1.000
.042	.162	.250	.080		.072	1.125
.048	.177		.092		.081	1.250
.063	.188		.106		.091	1.375
.072	.192		.120		.102	1.500
.080	.207		.125		.114	1.625
.090	.218		.135		.125	1.750
.107	.225		.148		.128	2.000
.130	.244		.156		.144	
.162	.250		.162		.156	
.177	.263		.177		.162	
	.283		.188		.182	
	.307		.192		.188	
	.313		.207		.250	
	.362		.218			
	.375		.225			
			.250			
			.312			
			.375			

Note 1. Square sizes of wire and bars are obtainable in preferred fractional sizes as follows: $\frac{1}{32}$, $\frac{1}{16}$, $\frac{3}{32}$, $\frac{1}{8}$, $\frac{5}{32}$, $\frac{3}{16}$, $\frac{1}{4}$, $\frac{5}{16}$, $\frac{3}{8}$, $\frac{7}{16}$, $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{4}$, $\frac{7}{8}$, $1\frac{1}{8}$, $1\frac{1}{4}$, $1\frac{3}{4}$, 2 in.

Note 2. Music Wire and Valve spring quality wire are available in round sections only.

Note 3. Music Wire is drawn to American Steel Wire & Music Wire Gage and to special gages. Steel wire is drawn to U.S. Steel Wire Gage (same as Washburn & Moen Gage). Copper and Nickel Alloys are drawn to the American Wire Gage (same as Brown & Sharpe Gage).